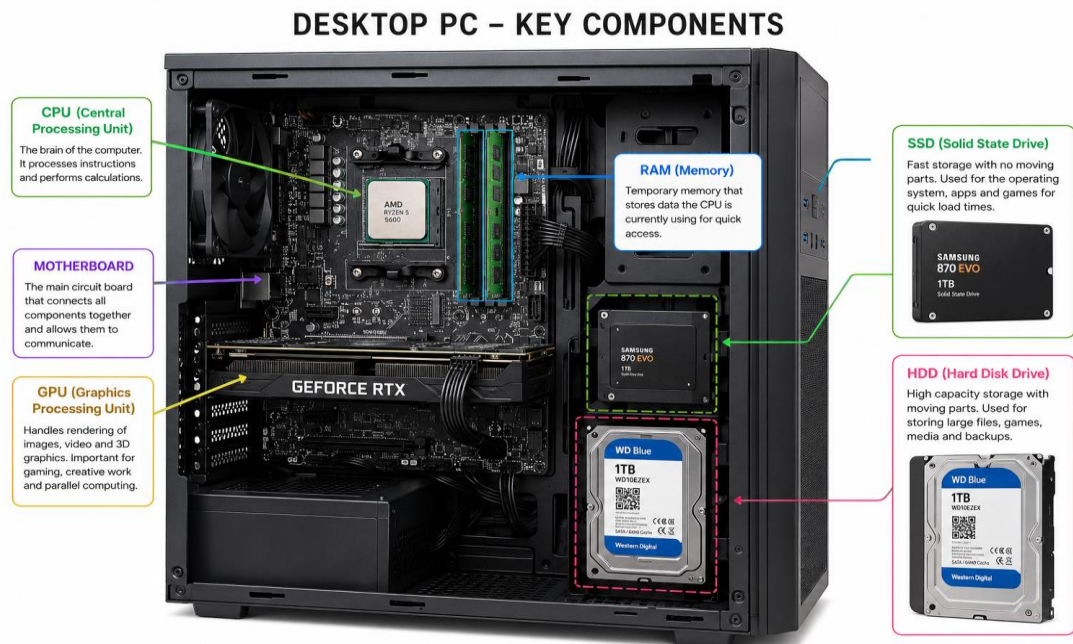


1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU and both HDD and SSD Storage Components.



2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability and typical use cases.

### Difference between HDD and SSD

#### HDD (Hard Disk Drive):-

- In case of file transfer and data access HDD is very Slower around 50 – 100 MB/s.
- HDD is very slow. So, it uses more power.
- In the booting of Operating system, it takes longer time.
- More vulnerable to physical shock.

#### SSD (Solid State Drive):-

- Much more faster in terms of file transfer and data access around 100 – 500 MB/s.
- SSD is using flash memory using with no moving parts. So, it reduce the risk of part failures.
- It is much faster in booting operating system. So, It can be used in gaming and streaming.
- More resistant to shock and vibration.

3. Write a short explanation (2-3 sentences each) describing the role of CPU, RAM and GPU in running a game like PUBG or Free Fire on a PC.

### **CPU (Central Processing Unit)**

- The CPU is the main processor of the computer and handles game logic, player actions, physics calculations, and communication between different hardware components. In games like PUBG or Free Fire, a faster CPU helps maintain smooth gameplay and reduces lag during complex actions.

### **RAM (Random Access Memory)**

- RAM temporarily stores active game data that the CPU and GPU need to access quickly, such as level textures, character models, and sound files. Having enough RAM ensures the game can load new areas smoothly without stuttering, preventing lag when parachuting into dense cities in PUBG.

### **GPU (Graphics Processing Unit)**

- The GPU is responsible for rendering every visual element you see on screen, from 3D character models and weapon skins to lighting, shadows, and environmental effects like smoke or grass. In fast-paced shooters, a strong GPU ensures high and stable frame rates (FPS), making aiming and spotting enemies smooth and responsive.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

### **Major Components needs to upgrade**

- GPU – Most important for resolution and visual settings.
- CPU – Critical for smooth gameplay
- RAM – To prevent the slow loading of new version of FIFA game.
- Storage(SSD) – It will eliminate the pop in textures when game loads.
- Power Supply Unit(PSU) – It enables the good power supply with upgrades.